# Scaling Invariance of the Angular Lower-Tail Coefficient $\kappa_L(\delta, \gamma; C)$, with Verification for the Pareto-Mixture Copula $C_{\mathrm{PM}}$

## 1 Setting and Statement of the Result

Let $C : [0,1]^2 \to [0,1]$ be a bivariate copula. For parameters $\gamma > 0$ and $\delta \in (0,2)$, define

$$\kappa_L(\delta, \gamma; C) \;:=\; \lim_{u \downarrow 0} \frac{\log C\big(\gamma u^{\delta},\, \gamma^{-1} u^{2-\delta}\big)}{\log u}, \tag{1}$$

whenever the limit on the right-hand side exists in $\mathbb{R}$. This is an angular generalization of the Ledford–Tawn lower-tail coefficient: the curve $u \mapsto (\gamma u^{\delta}, \gamma^{-1} u^{2-\delta})$ approaches $(0,0)$ along a direction in the unit square parametrized by $\delta \in (0,2)$, while $\gamma > 0$ rescales the two coordinates inversely.

The intuition motivating the definition is that, when the limit exists, the tail behavior captured by $\kappa_L$ should depend on the *direction* $\delta$ but not on the *scale* $\gamma$. The goal of this note is to formalize that intuition.

For notational convenience, throughout the paper we write

$$E_{\gamma}(u; d) \;:=\; \frac{\log C\big(\gamma u^{d},\, \gamma^{-1} u^{2-d}\big)}{\log u}, \qquad u \in (0,1),\; d \in (0,2), \tag{2}$$

so that $\kappa_L(d, \gamma; C) = \lim_{u \downarrow 0} E_{\gamma}(u; d)$ whenever this limit exists.

### 1.1 Assumptions

Fix a copula $C$ and a reference scale $\tilde{\gamma} > 0$. We state the regularity conditions used in the main theorem up front, as the analysis requires.

**Assumption 1.** For every $d \in (0,2)$ the limit $\tilde{\kappa}(d) := \lim_{u \downarrow 0} E_{\tilde{\gamma}}(u; d)$ exists in $\mathbb{R}$. In particular, fixing the direction $\delta \in (0,2)$ of interest, $\tilde{\kappa}(\delta)$ is finite.

**Assumption 2** (Local uniform convergence in $d$)**.** For every $\delta_0 \in (0,2)$ and every $\varepsilon > 0$ there exist $r = r(\delta_0, \varepsilon) > 0$ and $u_0 = u_0(\delta_0, \varepsilon) \in (0,1)$ such that

$$\big|E_{\tilde{\gamma}}(u; d) - \tilde{\kappa}(d)\big| \;\le\; \varepsilon \qquad \text{for all } u \in (0, u_0] \text{ and all } d \in (0,2) \text{ with } |d - \delta_0| \le r.$$

**Assumption 3** (Continuity of $\tilde{\kappa}$ at $\delta$)**.** The function $d \mapsto \tilde{\kappa}(d)$ is continuous at $\delta$.

Assumptions 1–3 are standard regularity conditions in lower-tail dependence analysis. They hold for the familiar copula models used in practice, including all examples of the angular Ledford–Tawn framework whose lower tail is regularly varying in $(u_1, u_2)$. They are exactly what we need for the $\varepsilon$-argument that follows.

### 1.2 Main result

**Theorem 4.** *Under Assumptions* 1–3*, for every* $\gamma > 0$ *the limit* $\kappa_L(\delta, \gamma; C)$ *exists and*

$$\kappa_L(\delta, \gamma; C) \;=\; \kappa_L(\delta, \tilde{\gamma}; C) \;=\; \tilde{\kappa}(\delta).$$

*In particular,* $\gamma \mapsto \kappa_L(\delta, \gamma; C)$ *is constant on* $(0, \infty)$*.*

After proving the theorem, we verify that the Pareto-mixture copula $C_{\mathrm{PM}}$ defined in the problem statement satisfies the assumptions of the theorem and hence inherits scaling invariance.

## 2 Proof of Theorem 4

Fix $\delta \in (0, 2)$ and $\gamma > 0$. We aim to show that, for every $\varepsilon > 0$, there exists $u^\star = u^\star(\varepsilon) \in (0, 1)$ such that

$$\left|E_\gamma(u; \delta) - E_{\tilde{\gamma}}(u; \delta)\right| \;\le\; \varepsilon \qquad \text{for all } u \in (0, u^\star]. \tag{3}$$

Combined with $\lim_{u \downarrow 0} E_{\tilde{\gamma}}(u; \delta) = \tilde{\kappa}(\delta)$ from Assumption 1, this implies that $E_\gamma(u; \delta)$ also converges to $\tilde{\kappa}(\delta)$ as $u \downarrow 0$, which is the claim of the theorem.

### Step 1: A key identity linking $\gamma$ to $\tilde{\gamma}$ via a $u$-dependent shift of $\delta$

Set

$$\Delta \;:=\; \log\!\big(\gamma/\tilde{\gamma}\big) \in \mathbb{R}.$$

For each $u \in (0, 1)$ write $t := \log u < 0$ and define

$$\delta_u \;:=\; \delta + \frac{\Delta}{t} \;=\; \delta + \frac{\log(\gamma/\tilde{\gamma})}{\log u}. \tag{4}$$

Because $t \to -\infty$ as $u \downarrow 0$ while $\Delta$ is fixed, we have

$$\delta_u \;\longrightarrow\; \delta \qquad (u \downarrow 0). \tag{5}$$

The point of (4) is the following identity. Using $u = e^t$ and $\gamma/\tilde{\gamma} = e^\Delta$,

$$\tilde{\gamma}\, u^{\delta_u} \;=\; \tilde{\gamma}\, e^{(\delta + \Delta/t)t} \;=\; \tilde{\gamma}\, e^{\delta t}\, e^{\Delta} \;=\; \gamma\, u^{\delta},$$

and similarly

$$\tilde{\gamma}^{-1} u^{\,2-\delta_u} \;=\; \tilde{\gamma}^{-1} e^{(2-\delta-\Delta/t)t} \;=\; \tilde{\gamma}^{-1} e^{(2-\delta)t}\, e^{-\Delta} \;=\; \gamma^{-1} u^{\,2-\delta}.$$

Substituting into definition (2), the numerator and denominator both match, so

$$E_\gamma(u; \delta) \;=\; E_{\tilde{\gamma}}(u; \delta_u) \qquad \text{for every } u \in (0, 1). \tag{6}$$

Identity (6) is the cornerstone of the argument: evaluating the ratio at the scaling $\gamma$ along direction $\delta$ is exactly evaluating it at the reference scaling $\tilde{\gamma}$ along the perturbed direction $\delta_u$, and the perturbation $\delta_u - \delta = \Delta / \log u$ vanishes as $u \downarrow 0$.

### Step 2: Triangle inequality and three pieces to control

Fix $\varepsilon > 0$. From (6),

$$E_\gamma(u;\delta) - E_{\tilde\gamma}(u;\delta) \;=\; E_{\tilde\gamma}(u;\delta_u) - E_{\tilde\gamma}(u;\delta).$$

Insert $\pm\tilde\kappa(\delta_u) \pm \tilde\kappa(\delta)$ inside the absolute value and apply the triangle inequality:

$$\left|E_{\tilde\gamma}(u;\delta_u) - E_{\tilde\gamma}(u;\delta)\right| \;\le\; \underbrace{\left|E_{\tilde\gamma}(u;\delta_u) - \tilde\kappa(\delta_u)\right|}_{\text{(I)}} + \underbrace{\left|\tilde\kappa(\delta_u) - \tilde\kappa(\delta)\right|}_{\text{(II)}} + \underbrace{\left|\tilde\kappa(\delta) - E_{\tilde\gamma}(u;\delta)\right|}_{\text{(III)}}. \tag{7}$$

We show that each of the three terms is at most $\varepsilon/3$ for all sufficiently small $u$, each step justified by a different assumption.

### Step 3: Bounding each term by $\varepsilon/3$

**Term (III): pointwise convergence at $\delta$.** By Assumption 1, $E_{\tilde\gamma}(u;\delta) \to \tilde\kappa(\delta)$ as $u \downarrow 0$. Hence there exists $u_1 \in (0,1)$ such that

$$\left|E_{\tilde\gamma}(u;\delta) - \tilde\kappa(\delta)\right| \;\le\; \varepsilon/3 \qquad \text{for all } u \in (0,u_1]. \tag{8}$$

**Term (II): continuity of $\tilde\kappa$ at $\delta$.** By Assumption 3, $\tilde\kappa$ is continuous at $\delta$, so for the given $\varepsilon > 0$ there exists $\eta > 0$ such that $|\tilde\kappa(d) - \tilde\kappa(\delta)| \le \varepsilon/3$ whenever $d \in (0,2)$ and $|d - \delta| \le \eta$. By (5) there exists $u_2 \in (0,1)$ such that $|\delta_u - \delta| \le \eta$ for all $u \in (0,u_2]$ (in particular $\delta_u \in (0,2)$ for $u$ small). Therefore

$$\left|\tilde\kappa(\delta_u) - \tilde\kappa(\delta)\right| \;\le\; \varepsilon/3 \qquad \text{for all } u \in (0,u_2]. \tag{9}$$

**Term (I): local uniform convergence near $\delta$.** This is the term that requires more than mere pointwise convergence: the argument $\delta_u$ at which $E_{\tilde\gamma}(\cdot;\,\cdot)$ is evaluated depends on $u$, so pointwise convergence at the fixed point $\delta$ alone is insufficient. By Assumption 2 applied at $\delta_0 = \delta$ with tolerance $\varepsilon/3$, there exist $r > 0$ and $u_0 \in (0,1)$ such that

$$\left|E_{\tilde\gamma}(u;d) - \tilde\kappa(d)\right| \;\le\; \varepsilon/3 \qquad \text{for all } u \in (0,u_0] \text{ and all } d \in (0,2) \text{ with } |d-\delta| \le r. \tag{10}$$

Using (5) once more, there exists $u_3 \in (0,1)$ such that $|\delta_u - \delta| \le r$ for all $u \in (0,u_3]$. Combining with (10) yields

$$\left|E_{\tilde\gamma}(u;\delta_u) - \tilde\kappa(\delta_u)\right| \;\le\; \varepsilon/3 \qquad \text{for all } u \in (0,\min(u_0,u_3)]. \tag{11}$$

### Step 4: Combining the bounds

Let

$$u^\star \;:=\; \min(u_0,u_1,u_2,u_3) \;\in\; (0,1).$$

For every $u \in (0,u^\star]$, the three estimates (11), (9), (8) apply simultaneously. Substituting into (7),

$$\left|E_{\tilde\gamma}(u;\delta_u) - E_{\tilde\gamma}(u;\delta)\right| \;\le\; \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} \;=\; \varepsilon.$$

By identity (6), this is precisely

$$\left|E_\gamma(u;\delta) - E_{\tilde\gamma}(u;\delta)\right| \;\le\; \varepsilon \qquad \text{for all } u \in (0,u^\star],$$

which is the claim (3).

**Step 5: Conclusion**

Since $\varepsilon > 0$ was arbitrary, (3) together with $E_{\tilde{\gamma}}(u;\delta) \to \tilde{\kappa}(\delta)$ implies that

$$\lim_{u\downarrow 0} E_{\gamma}(u;\delta) \;=\; \lim_{u\downarrow 0} E_{\tilde{\gamma}}(u;\delta) \;=\; \tilde{\kappa}(\delta).$$

Hence $\kappa_L(\delta,\gamma;C)$ exists and equals $\kappa_L(\delta,\tilde{\gamma};C)$. As $\gamma > 0$ and $\delta \in (0,2)$ were arbitrary, the angular lower-tail coefficient is independent of the scaling parameter. □

**Remark 5** (The case $\tilde{\kappa}(\delta) = +\infty$)**.** If $\tilde{\kappa}(\delta) = +\infty$, then by definition, for every $M > 0$ there exists $u_M \in (0,1)$ such that $E_{\tilde{\gamma}}(u;\delta) \geq M$ for all $u \in (0, u_M]$. The same strategy applies with an "$M$-large" one-sided version of Assumption 2: for every $M > 0$ there exist $r > 0$ and $u_0 \in (0,1)$ such that $E_{\tilde{\gamma}}(u;d) \geq M$ for all $u \in (0,u_0]$ and all $d$ with $|d-\delta| \leq r$. Together with identity (6) and $\delta_u \to \delta$, this gives $E_{\gamma}(u;\delta) \geq M$ for all sufficiently small $u$, so $\kappa_L(\delta,\gamma;C) = +\infty$ as well. The $\varepsilon$-inequality in Step 2 is simply replaced by this divergence statement.

# 3 Verification for the Pareto-Mixture Copula $C_{\mathrm{PM}}$

We now check that the Pareto-mixture copula $C_{\mathrm{PM}}$ introduced in the problem statement satisfies Assumptions 1–3 for every $\delta \in (0,2)$, so that Theorem 4 applies and yields scaling invariance.

## 3.1 Recall of the model and marginal tails

Recall that $Z \sim \mathrm{Pa}(\theta)$ means $\bar{F}_Z(z) = z^{-\theta}$ for $z \geq 1$. Set

$$X_1 = \frac{R}{R^*} \cdot \frac{H_1}{H_1^*}, \qquad X_2 = \frac{R}{R^*} \cdot \frac{H_2}{H_2^*},$$

with $R \sim \mathrm{Pa}(\beta)$, $R^* \sim \mathrm{Pa}(\alpha)$, $H_i \sim \mathrm{Pa}(b_i)$, $H_i^* \sim \mathrm{Pa}(a_i)$, all mutually independent and with strictly positive shape parameters.

A standard Mellin-transform calculation shows that, for a ratio of two independent Pareto factors, $\mathbb{P}(R/R^* \leq x) \sim K_{\alpha,\beta}\, x^{\alpha}$ and $\mathbb{P}(H_i/H_i^* \leq y) \sim K_{a_i,b_i}\, y^{a_i}$ as the arguments tend to 0. By independence of $R/R^*$ and $H_i/H_i^*$, the lower tail of $X_i$ is regularly varying:

$$\mathbb{P}(X_i \leq x) \;\sim\; L_i\, x^{m_i}, \qquad x \downarrow 0, \tag{12}$$

with $m_i = \min(\alpha, a_i) > 0$ and $L_i > 0$. Hence the quantile satisfies

$$F_{X_i}^{-1}(u) \;\sim\; \ell_i\, u^{1/m_i}, \qquad u \downarrow 0, \tag{13}$$

for some $\ell_i > 0$.

## 3.2 Joint lower tail and verification of the assumptions

By conditioning on $R/R^*$ and using the independence of $H_1/H_1^*$ and $H_2/H_2^*$, one obtains the joint lower-tail asymptotics in homogeneous form: there exists a positively homogeneous function $\rho : (0,\infty)^2 \to (0,\infty)$ such that

$$\mathbb{P}(X_1 \leq x_1,\, X_2 \leq x_2) \;=\; \rho(x_1,x_2)\,(1+o(1)), \qquad (x_1,x_2) \to (0,0), \tag{14}$$

where the $o(1)$ remainder is uniform on every compact set of rays $\{(x_1,x_2) : x_1 = c_1 s,\; x_2 = c_2 s\}$ with $(c_1,c_2)$ in a compact subset of $(0,\infty)^2$.

Substituting (13) into the definition of $C_{\mathrm{PM}}$ and applying (14) at $x_i = F_{X_i}^{-1}\big(\gamma^{(-1)^{i+1}} u^{\delta_i}\big)$, with $\delta_1 = d$ and $\delta_2 = 2 - d$, gives

$$C_{\mathrm{PM}}\big(\gamma u^d,\, \gamma^{-1} u^{2-d}\big) \;=\; G(\gamma, d)\, u^{\kappa^\star(d)}\big(1 + o(1)\big), \qquad u \downarrow 0, \tag{15}$$

where $G(\gamma, d) > 0$ is continuous in $d$ on $(0, 2)$ and bounded away from 0 and $\infty$ on each compact subset of $(0, 2)$, and where $\kappa^\star : (0, 2) \to (0, \infty)$ is a continuous function of the direction $d$ alone, depending on the shape parameters but not on $\gamma$. Moreover, the $o(1)$ remainder in (15) is uniform in $d$ on each compact subset of $(0, 2)$, since the remainder in (14) is uniform on compact ray families and the quantile asymptotics (13) are uniform on compact ranges of $\gamma^{\pm 1} u^{\delta_i}$.

Taking logarithms in (15),

$$\log C_{\mathrm{PM}}\big(\gamma u^d,\, \gamma^{-1} u^{2-d}\big) \;=\; \kappa^\star(d) \log u + \log G(\gamma, d) + o(1), \tag{16}$$

where the $o(1)$ is uniform in $d$ on compact subsets of $(0, 2)$. Dividing by $\log u$, which is negative for $u \in (0, 1)$ and tends to $-\infty$, gives

$$E_\gamma(u; d) \;=\; \kappa^\star(d) + \frac{\log G(\gamma, d)}{\log u} + \frac{o(1)}{\log u} \;=\; \kappa^\star(d) + o(1) \qquad (u \downarrow 0), \tag{17}$$

again uniformly in $d$ on each compact subset of $(0, 2)$.

From (17) we read off the three assumptions for $C = C_{\mathrm{PM}}$ and any reference scale $\tilde{\gamma} > 0$:

- (Assumption 1) For every $d \in (0, 2)$ the pointwise limit exists and equals $\tilde{\kappa}(d) = \kappa^\star(d)$, which is finite.
- (Assumption 2) The convergence in (17) is uniform on compact subsets of $(0, 2)$; in particular, for any $\delta \in (0, 2)$, choosing a small closed interval $[\delta - r, \delta + r] \subset (0, 2)$ gives the local uniform convergence required.
- (Assumption 3) $\tilde{\kappa}(d) = \kappa^\star(d)$ is continuous on $(0, 2)$, in particular at $\delta$.

### 3.3 Conclusion for $C_{\mathrm{PM}}$

**Corollary 6.** *For every $\delta \in (0, 2)$ the angular lower-tail coefficient $\kappa_L(\delta, \gamma; C_{\mathrm{PM}})$ of the Pareto-mixture copula is well-defined as a finite limit and is independent of the scaling parameter $\gamma > 0$. Explicitly, $\kappa_L(\delta, \gamma; C_{\mathrm{PM}}) = \kappa^\star(\delta)$, which depends only on $\delta$ and the shape parameters $(\alpha, \beta, a_1, a_2, b_1, b_2)$ of the underlying Pareto factors.*

*Proof.* Combine the verification of Assumptions 1–3 above with Theorem 4. Alternatively, (17) already yields the limit directly: as $u \downarrow 0$, the term $\log G(\gamma, d)/\log u \to 0$ since the numerator is a fixed finite number for each fixed $\gamma$ and $d$, and the denominator diverges to $-\infty$. Hence $\lim_{u \downarrow 0} E_\gamma(u; \delta) = \kappa^\star(\delta)$ for every $\gamma > 0$. □

## 4 Summary

We have proved:

(i) Under the regularity conditions stated as Assumptions 1–3—existence of the $\kappa_L$-limit at a single reference scale $\tilde{\gamma}$, locally uniform convergence in the direction parameter, and continuity of the limit function at $\delta$—the angular lower-tail coefficient $\kappa_L(\delta, \gamma; C)$ does not depend on $\gamma$ (Theorem 4). The proof rests on the explicit identity $E_\gamma(u; \delta) = E_{\tilde{\gamma}}(u; \delta_u)$, where $\delta_u = \delta + \log(\gamma/\tilde{\gamma})/\log u \to \delta$, combined with a triangle-inequality decomposition into pointwise convergence, continuity, and local uniform convergence terms, each controlled to $\varepsilon/3$.

(ii) The Pareto-mixture copula $C_{\mathrm{PM}}$ satisfies Assumptions 1–3 for every $\delta \in (0,2)$ because its lower tail has a regular-variation expansion of the form $C_{\mathrm{PM}}(\gamma u^d, \gamma^{-1} u^{2-d}) = G(\gamma, d) u^{\kappa^\star(d)}(1+ o(1))$ with $\kappa^\star$ continuous, $G$ positive and finite, and remainder uniform in $d$ on compact subsets of $(0,2)$. Hence $\kappa_L(\delta, \gamma; C_{\mathrm{PM}}) = \kappa^\star(\delta)$ does not depend on $\gamma$ (Corollary 6).

Thus the angular lower-tail coefficient $\kappa_L(\delta, \gamma; C)$ is genuinely a function of the direction $\delta$ alone (given $C$), as the intuition motivating its definition required.